In contemplating ways to eliminate the many worlds implied by a bare-bones version of the underlying quantum formalism, we have explored chopping off the worlds by a random event (GRW) or reaching some kind of threshold (Penrose) or picking out particular worlds as real by adding additional variables (de Broglie–Bohm). What's left?

The problem is that the appearance of multiple branches of the wave function is automatic once we believe in wave functions and the Schrödinger equation. So the alternatives we have considered thus far either eliminate those branches or posit something that picks out one of them as special.

A third way suggests itself: deny the reality of the wave function entirely.

By this we don't mean to deny the central importance of wave functions in quantum mechanics. Rather, we can use wave functions, but we might not claim that they represent part of reality. They might simply characterize our knowledge; in particular, the incomplete knowledge we have about the outcome of future quantum measurements. This is known as the "epistemic" approach to quantum mechanics, as it thinks of wave functions as capturing something about what we know, as opposed to "ontological" approaches that treat the wave function as describing objective reality. Since wave functions are usually denoted by the Greek letter Ψ (Psi), advocates of epistemic approaches to quantum mechanics sometimes tease Everettians and other wave-function-realists by calling them "Psi-ontologists."

We've already noted that an epistemic strategy cannot work in the most naïve and straightforward way. The wave function is not a probability distribution; real probability distributions are never negative, so they can't lead to interference phenomena such as we observe in the double-slit experiment. Rather than giving up, however, we can try to be a bit more sophisticated in how we think about the relationship between the wave function and the real world. We can imagine building up a formalism that allows us to use wave functions to calculate the probabilities associated with experimental outcomes, while not attaching any underlying reality to them. This is the task taken up by epistemic approaches.

There have been many attempts to interpret the wave function epistemically, just as there are competing collapse models or hidden-variable theories. One of the most prominent is Quantum Bayesianism,